

Erdős Problem 915: Cheat Sheet (front)

one line per bound, page numbers into main.pdf
conjectural

· **PROVED** exact value established
OPEN general value open

PARTIAL partial or

Simple graphs

$\ell_m(n)$ edge

PROVED

UP $\left\lfloor \frac{m(n-1)}{2} \right\rfloor$, every n : Gomory–Hu tree, at most $n-1$ edges cross one tree cut and the rest cross two, against total cut capacity $(m-1)(n-1)$ [p. 36–39]

LO $\left\lfloor \frac{m(n-1)}{2} \right\rfloor$ $n \geq m \geq 2$: hub-and-spokes graph, spoke degree $\leq m-2$ [p. 40]

$k_m(n)$ vertex

PARTIAL

UP $= \left\lfloor \frac{m(n-1)}{2} \right\rfloor$ $m \leq 4, n \geq m$: the edge bound does not transfer ($\kappa \leq \lambda$ is the wrong direction), equal anyway up to $m=4$ (Bárfai, Bollobás, Leonard) [p. 6]

LO $\left\lfloor \frac{m(n-1)}{2} \right\rfloor$ $n \geq m \geq 2$: same hub-and-spokes [p. 40], and Sørensen–Thomassen’s value $\lfloor 8n/3 \rfloor - 4$ overtakes it at $m=5, n \geq 14$ [p. 6]

$\ell_m^{\text{dir}}(n)$ arc

OPEN

UP $M(n) = \max(2(n-1), \lfloor n^2/4 \rfloor)$ $m=2$: a feasible digraph is its own reachability skeleton, and a skeleton has at most $M(n)$ arcs [p. 47,51]

LO $M(n)$: bidirected hub [p. 45] or one-way bipartite digraph [p. 46]

$k_m^{\text{dir}}(n)$ vertex

OPEN

UP $M(n)$ $m=2$: an arc outside the skeleton has a second route, internally disjoint from the arc itself [p. 47,50]

LO $\lfloor n^2/4 \rfloor$: one-way bipartite digraph [p. 46]

Multigraphs

$L_m(n)$ edge

PROVED

UP $(m-1)(n-1)$, every n, m : Gomory–Hu, $n-1$ cuts each of capacity $\leq m-1$, every edge crosses one [p. 39]

LO $(m-1)(n-1)$: any spanning tree at multiplicity $m-1$ [p. 39]

$K_m(n)$ vertex

PARTIAL

UP $2(n-1)$ $m=3$: incidence-rank lemma at $r=2$ [p. 70,75]

LO $(m-1)(n-1)$: tree thickened to multiplicity $m-1$, no detours, so the whole cap goes to multiplicity [p. 75]

$L_m^{\text{dir}}(n)$ arc

PROVED

UP $(m-1)M(n)$, every n, m : peel a reachability skeleton off $m-1$ times [p. 58–59]

LO $(m-1)M(n)$: thickened bidirected tree or one-way bipartite digraph [p. 57]

$K_m^{\text{dir}}(n)$ vertex

OPEN

UP $(m-1) \lfloor n^2/4 \rfloor + 4(m-1)^2(n-1)$ far from tight at small n [p. 83]

LO $n(n-1)(m+1-n)$ proposed, $2 \leq n \leq m$: thickened complete digraph, 24 at $n=3, m=6$ [p. 83]

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Hypergraphs

$\ell_m^{(r)}(n)$ **edge**

PARTIAL

UP $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$: cut induction on trace hypergraphs (Berge forbids the graph reduction) [p. 64]
LO $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$ $m-1 \leq \binom{n-2}{r-2}$: star hypertree or degree-capped hub [p. 65–66]

$k_m^{(r)}(n)$ **vertex**

PARTIAL

UP $\left\lfloor \frac{n-1}{r-1} \right\rfloor$ $m=2$: incidence graph is a forest ($r|E| \leq n + |E| - 1$) [p. 67]
LO $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$ $r \geq 3, m-1 \leq \binom{n-2}{r-2}$: degree-capped hub [p. 67]

$\ell_m^{\text{dir}(r)}(n)$ **arc**

OPEN

UP $\frac{(m-1)n(n-1)}{r-1}$ not tight for large n : tail–head pair counting [p. 84]
LO $n \cdot \min(m-1, \binom{n-1}{r-1})$ $n \geq r$: bounded outgoing hyperedge count [p. 85]

$k_m^{\text{dir}(r)}(n)$ **vertex**

OPEN

UP $\frac{(m-1)n(n-1)}{r-1}$: same pair counting, one-step routes are also internally disjoint [p. 86]
LO $n \cdot \min(m-1, \binom{n-1}{r-1})$ $n \geq r$: same construction, the outgoing cut ignores interior vertices [p. 85]

Multihypergraphs

$L_m^{(r)}(n)$ **edge**

PARTIAL

UP $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$, unchanged by repetition: same cut induction, repetition-blind [p. 64]
LO $\frac{(m-1)(n-1)}{r-1}$ $(r-1) \mid (n-1)$: star hypertree, every hyperedge repeated $m-1$ times [p. 65]

$K_m^{(r)}(n)$ **vertex**

PARTIAL

UP $\left\lfloor \frac{2(n-1)}{r-1} \right\rfloor$ $m=3$: incidence-graph cycle rank via cut vertices and separating pairs [p. 70]
LO $\frac{(m-1)(n-1)}{r-1}$ $(r-1) \mid (n-1)$: repeated star hypertree [p. 69]

$L_m^{\text{dir}(r)}(n)$ **arc**

OPEN

UP $\frac{(m-1)n(n-1)}{r-1}$: tail–head pair counting [p. 84]
LO $n(m-1)$ $n \geq r$: every tail gets $m-1$ copies, meeting the bound exactly at $n=r$ [p. 85]

$K_m^{\text{dir}(r)}(n)$ **vertex**

OPEN

UP $\frac{(m-1)n(n-1)}{r-1}$: carries over from the arc problem, and the general (any tail/head split) orientation shares the constant [p. 84,86,90]
LO $n(m-1)$ $n \geq r$: same construction [p. 85]